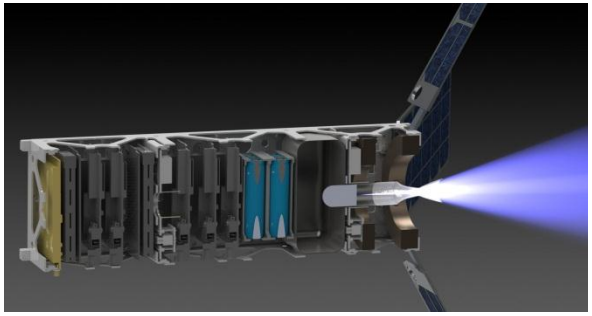
A Large Delta-V Plasma Thruster for Nanosatellites

B.W. Longmier and J.P. Sheehan
University of Michigan
Aerospace Engineering

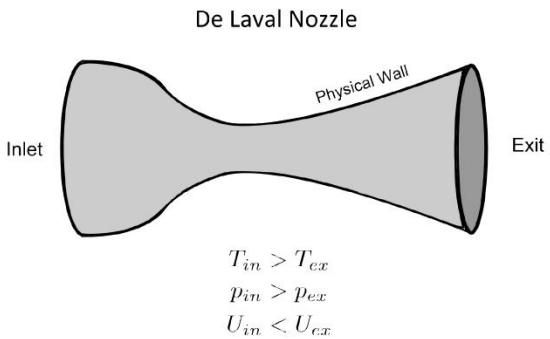
Outline

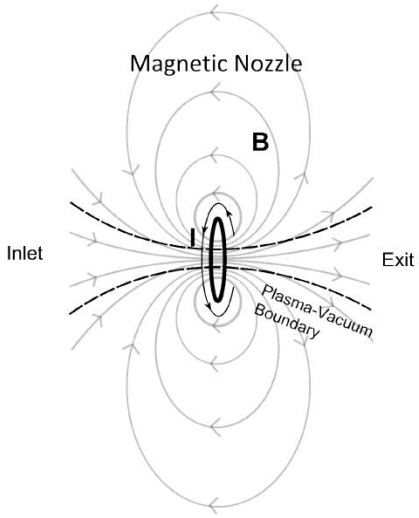
- Magnetic nozzles
- Ambipolar acceleration physics
- CubeSat Ambipolar Thruster (CAT) design
- Expected performance parameters
- Newly enabled missions



Magnetic nozzles guide plasma like physical nozzles guide fluids

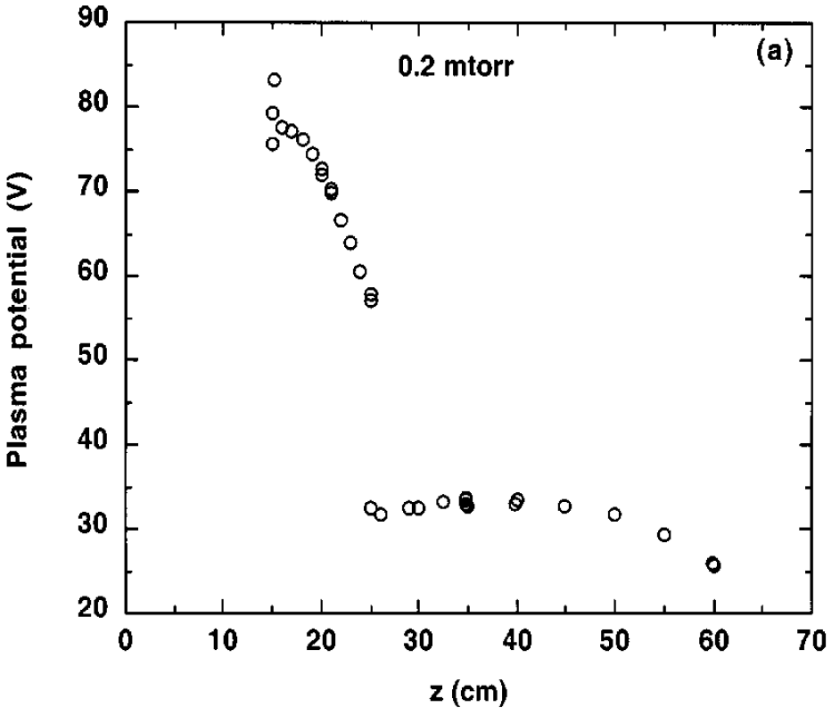
- Solenoid generates converging-diverging magnetic field
- Sufficiently high fields trap particles on field lines
- Plasma accelerated to sound speed at throat, further accelerated in diverging region
- Detachment occurs downstream





F. Ebersohn, B.W. Longmier

Current-free double layers (CFDLs) are often observed in diverging region

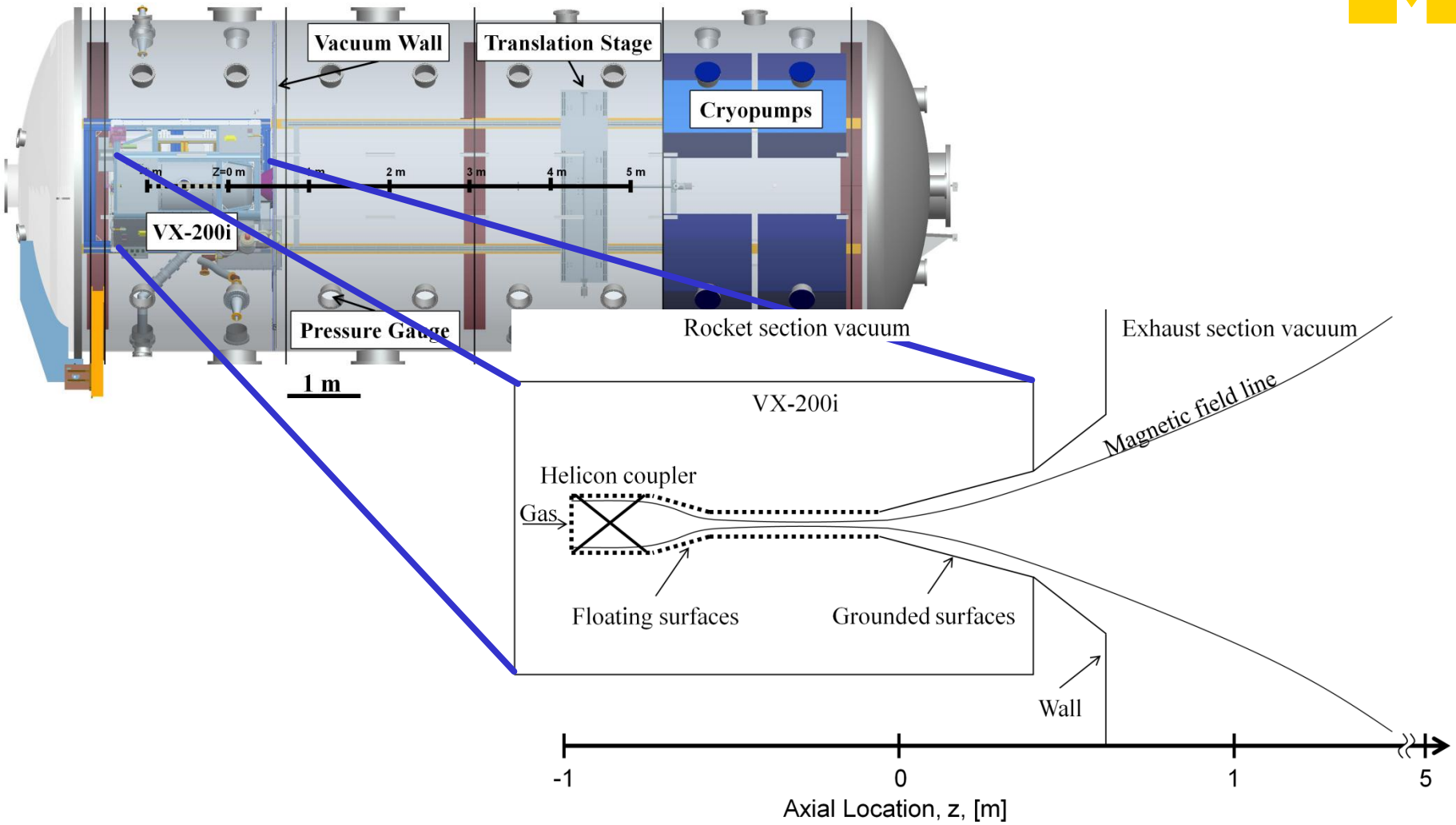


C. Charles and R. Boswell, "Current-free double-layer formation in a high-density helicon discharge," *Applied Physics Letters,* vol. 82, pp. 1356-1358, Mar 2003.

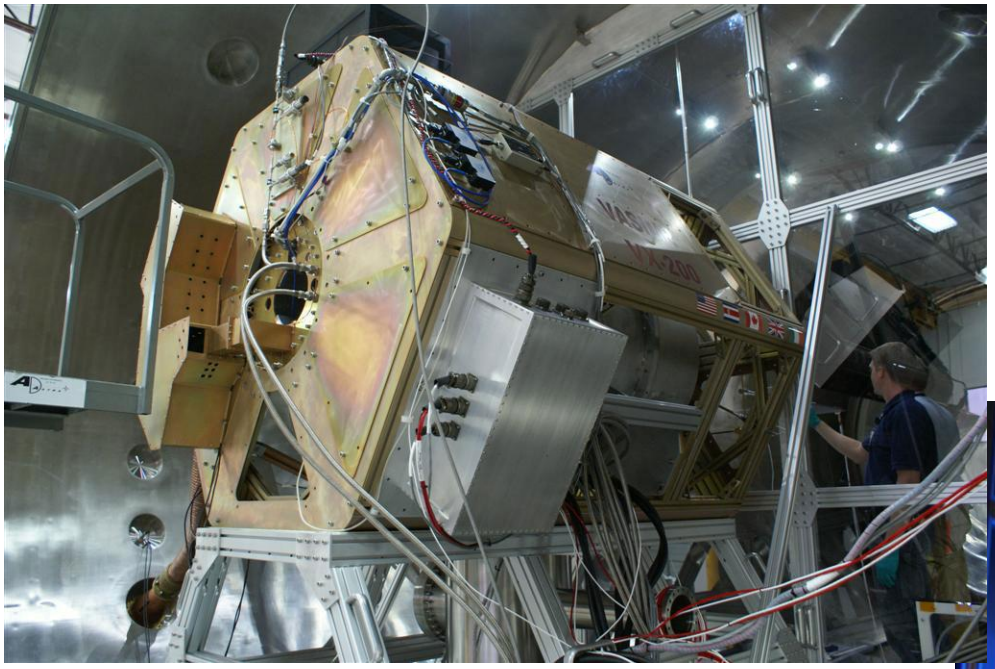
- Potential drop of $>T_e$ over distance of 10s of λ_d
- Accelerates ions away from nozzle
- Only exists for sufficiently low neutral pressures (typically < 1 mTorr)
- Upstream electron temperature ~25% larger than downstream
- Potential drop well described by Boltzmann relation (constant T_e)

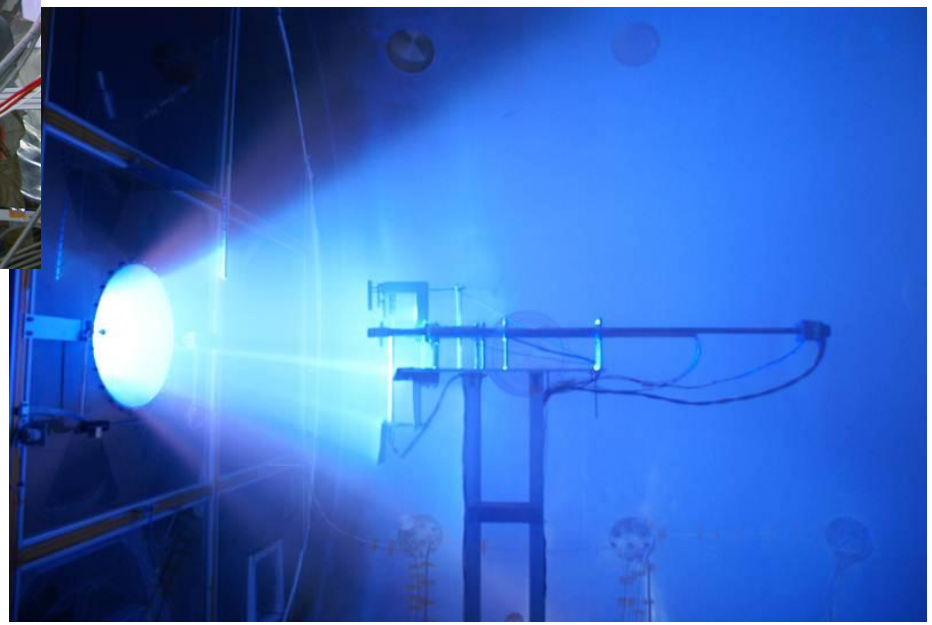
$$n_e = n_0 \exp\left(\frac{-e\varphi}{T_e}\right)$$

Experiments were performed using only the helicon stage of VASIMR

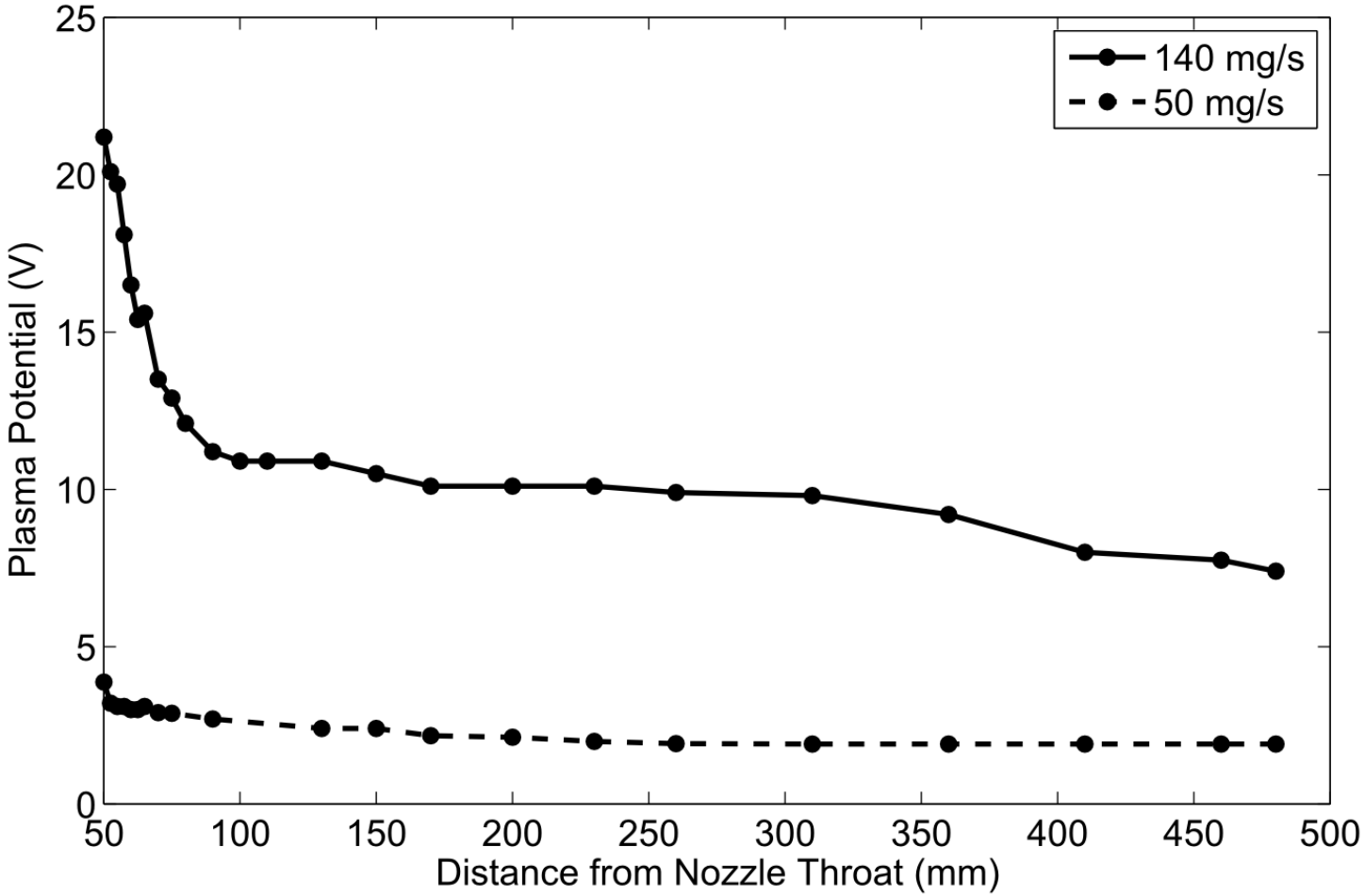


VASIMR

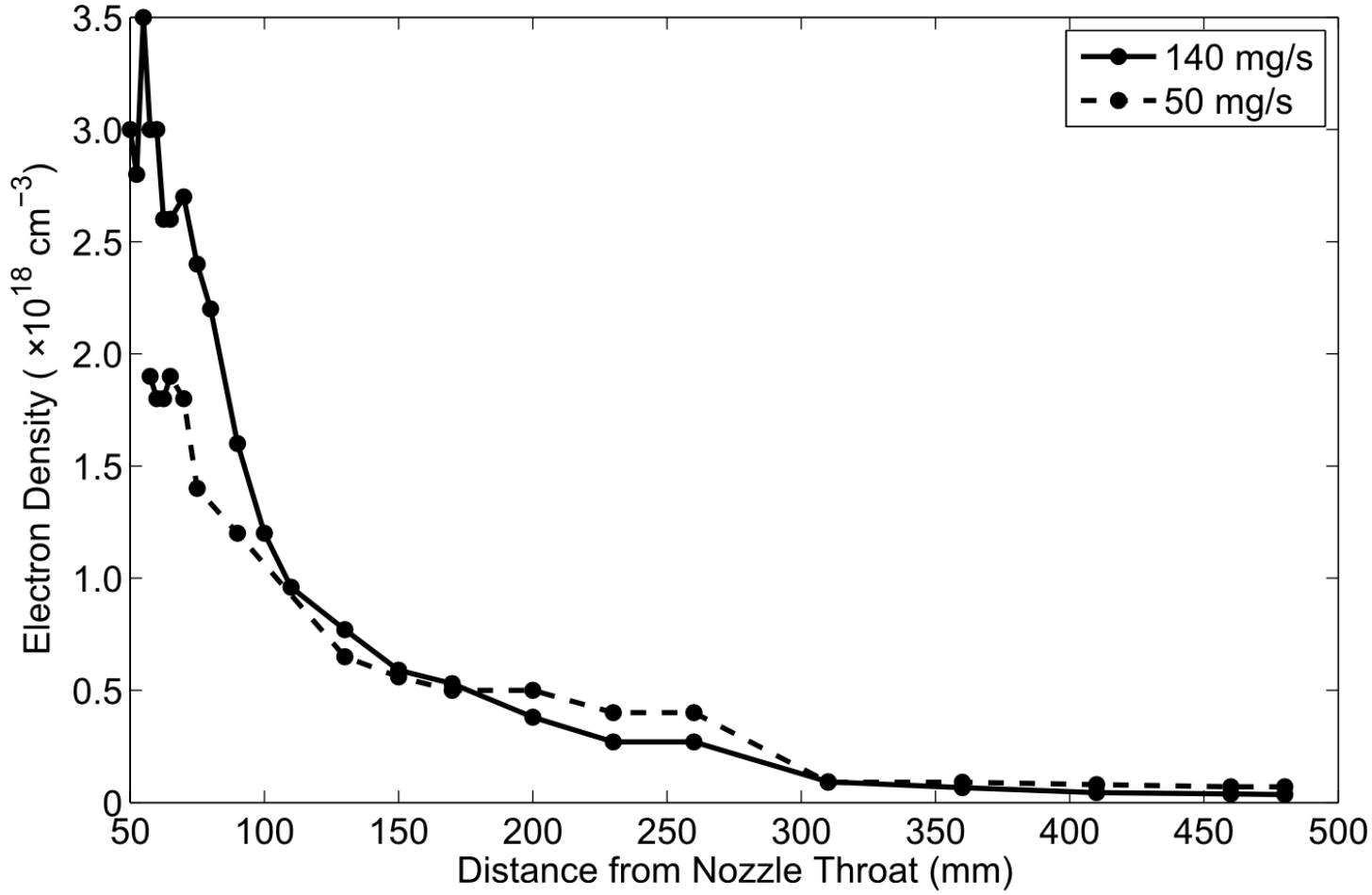




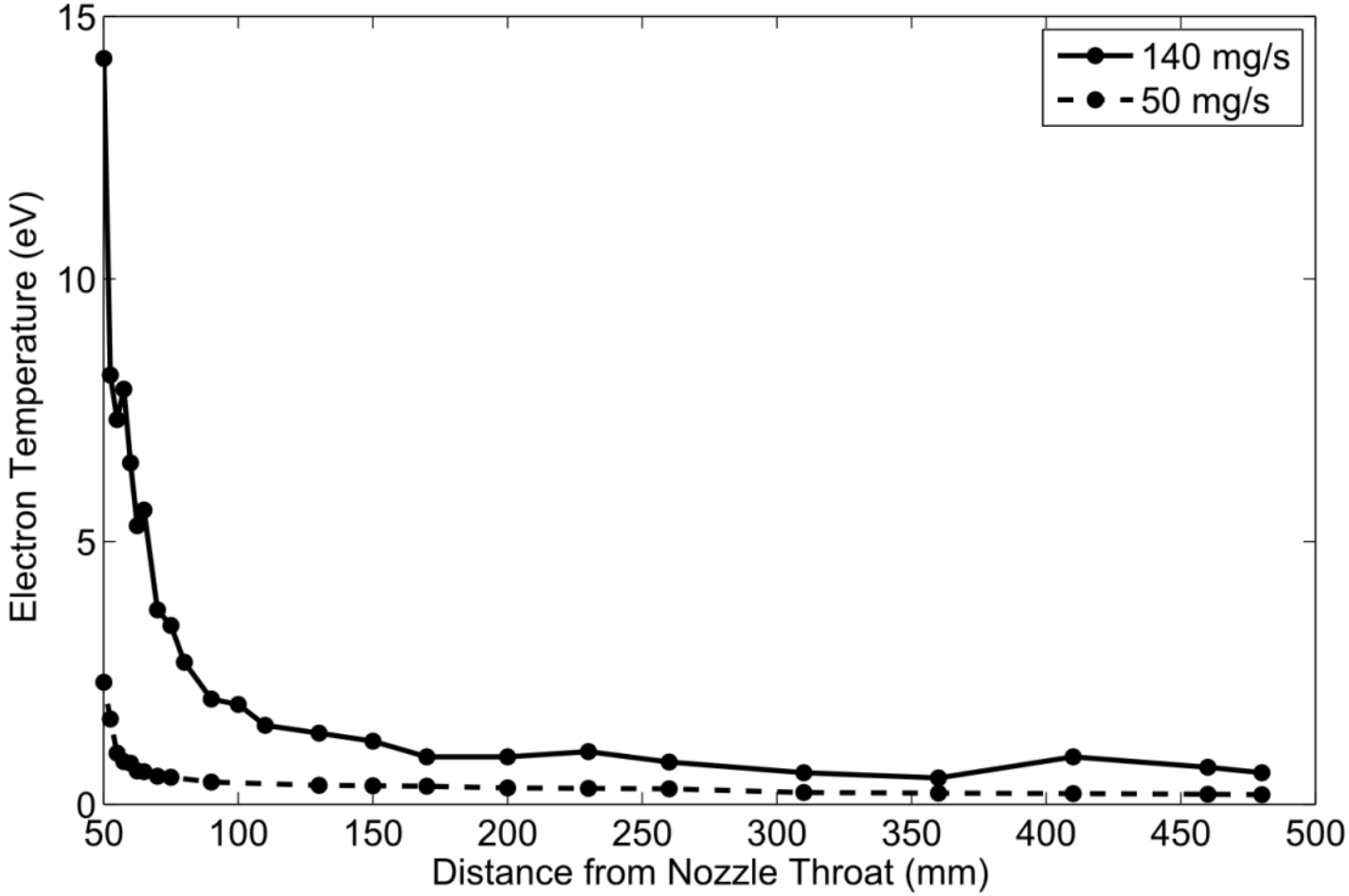
Plasma potential drops over 1000s of Debye lengths



Electron density drops on scale of potential structure



Large variation in electron temperature on scale of potential structure



Electron temperature variation significantly affects potential structure

Fluid Theory

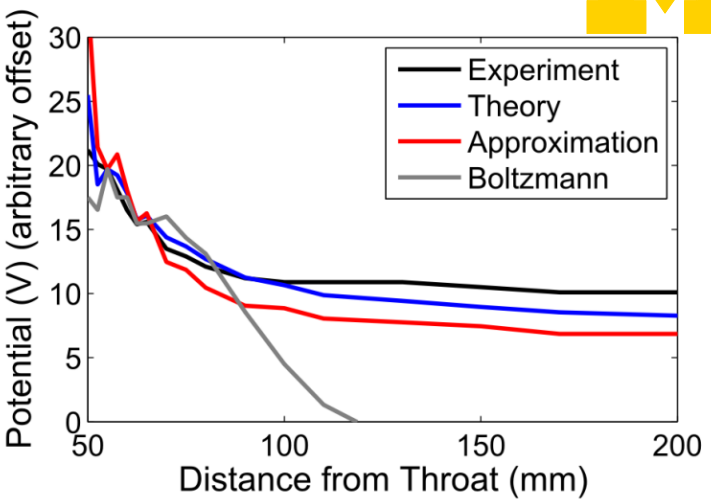
$$\frac{d\varphi}{ds} = \frac{1}{n_e}\frac{d(n_e T_e)}{ds}$$

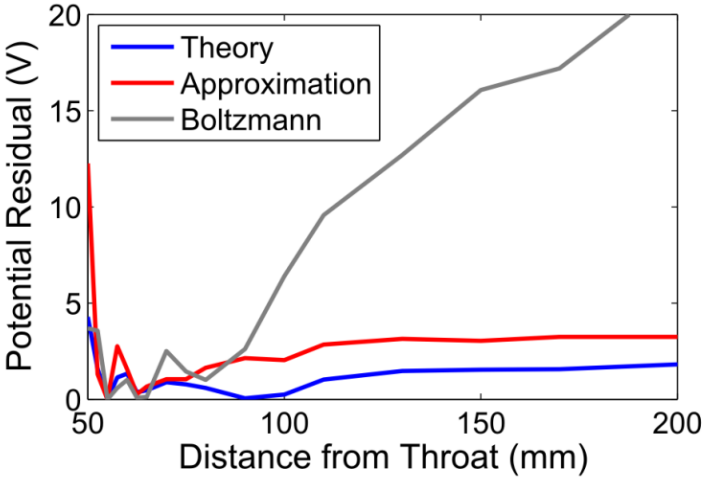
Approximation

$$\Delta\varphi = 2\Delta T_e$$

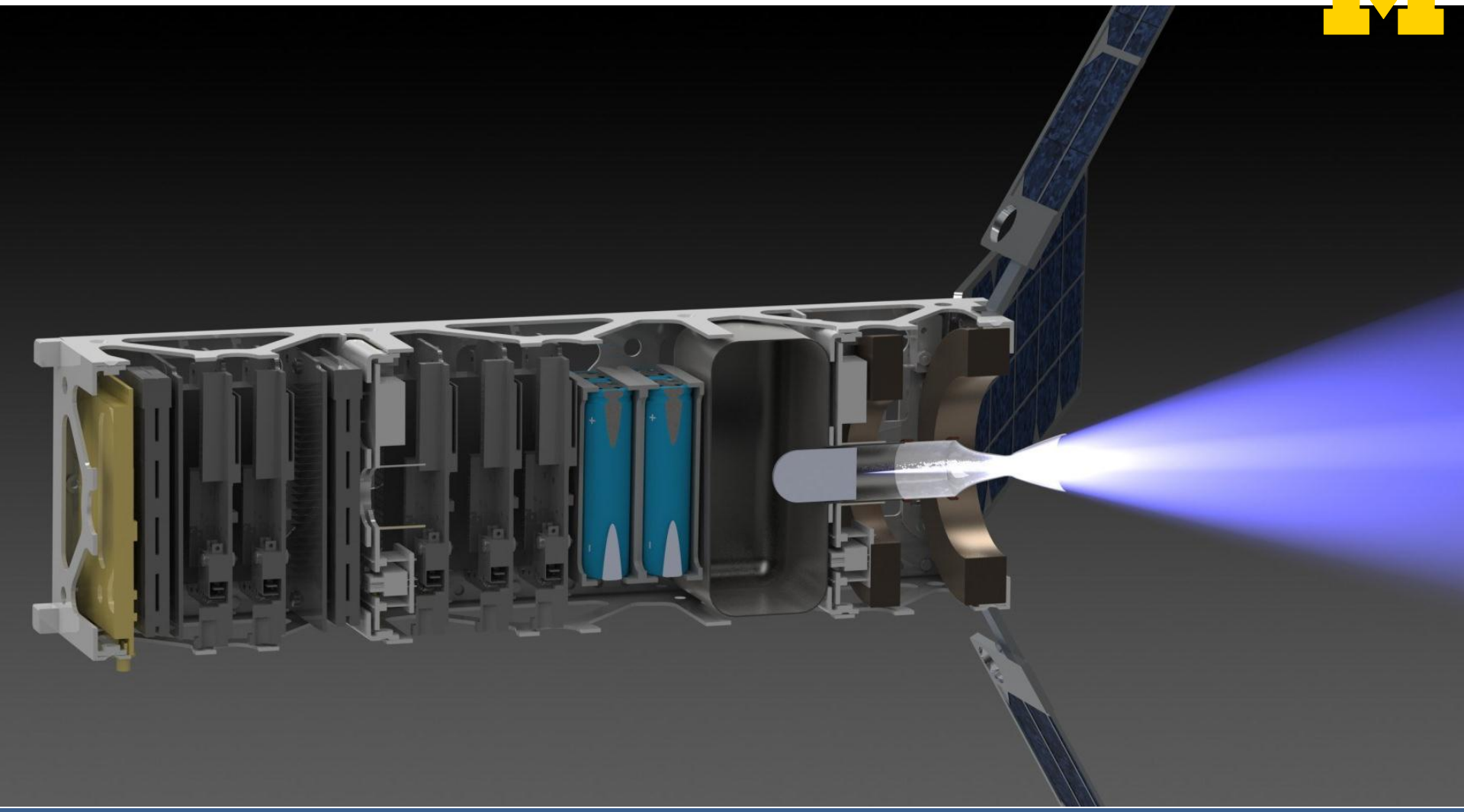
Boltzmann Relation

$$\Delta\varphi = T_{e,i}\ln\left(\frac{n_{e,f}}{n_{e,i}}\right)$$

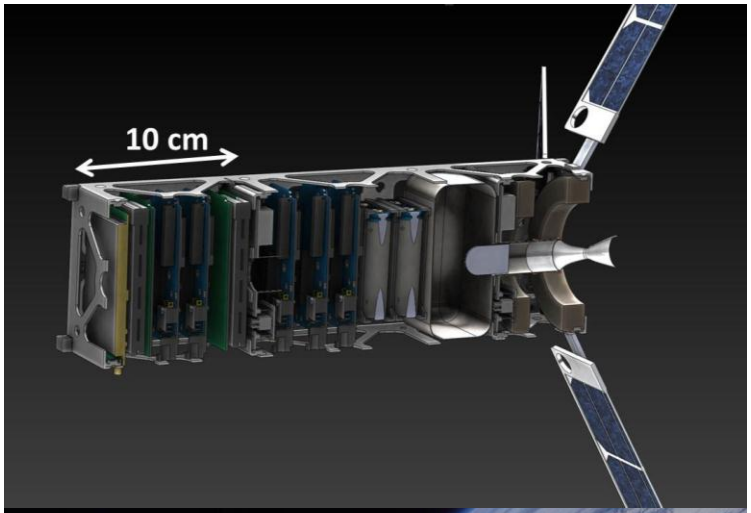




CubeSat Ambipolar Thruster



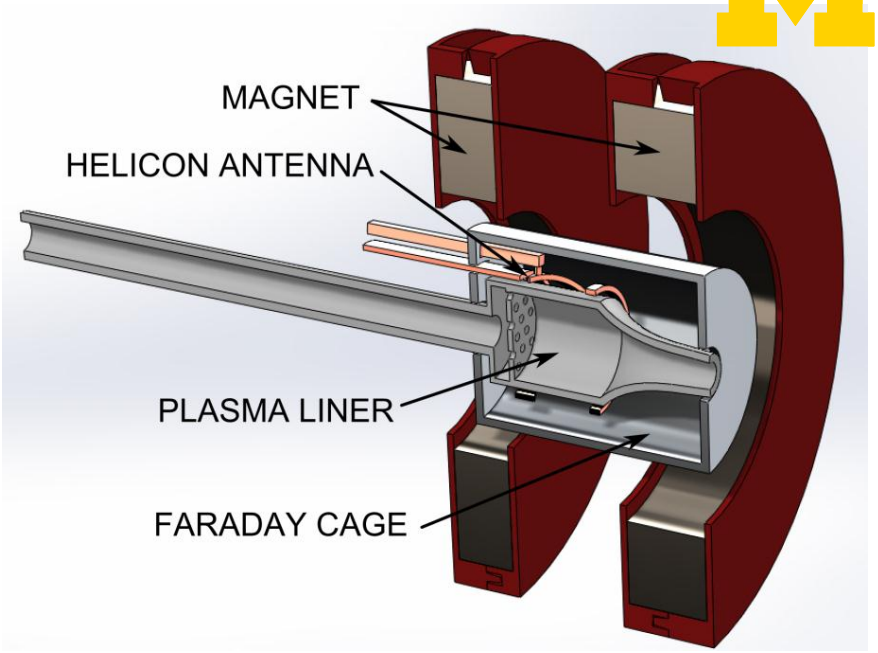
CubeSat Ambipolar Thruster will allow CubeSats to make large ΔV maneuvers

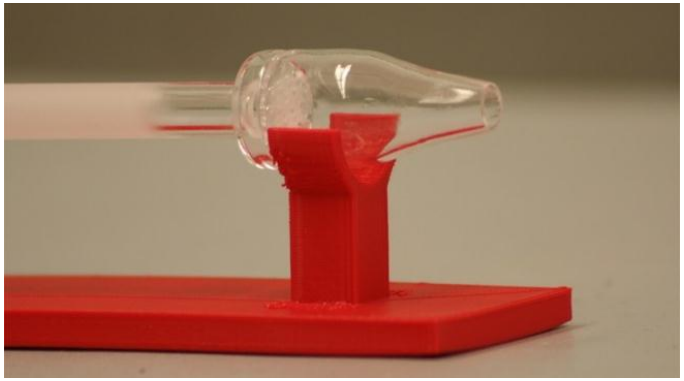


- CubeSat Ambipolar Thruster (CAT) designed to fit CubeSat form factor
- Can provide thrust for:
  - Orbit adjustment, formation flying
  - Earth escape + 3km/s remaining delta-V
  - Low altitude flying (180-200km for 10-100 min)
- Liquid storable propellants used for:
  - ΔV of 10 km/s possible
  - Higher atomic mass propellants for large thrust/power ratio
  - Lower atomic mass propellants for higher I_{sp}
  - Lower ionization cost propellants for higher thruster efficiency

CAT design uses a helicon plasma and a magnetic nozzle

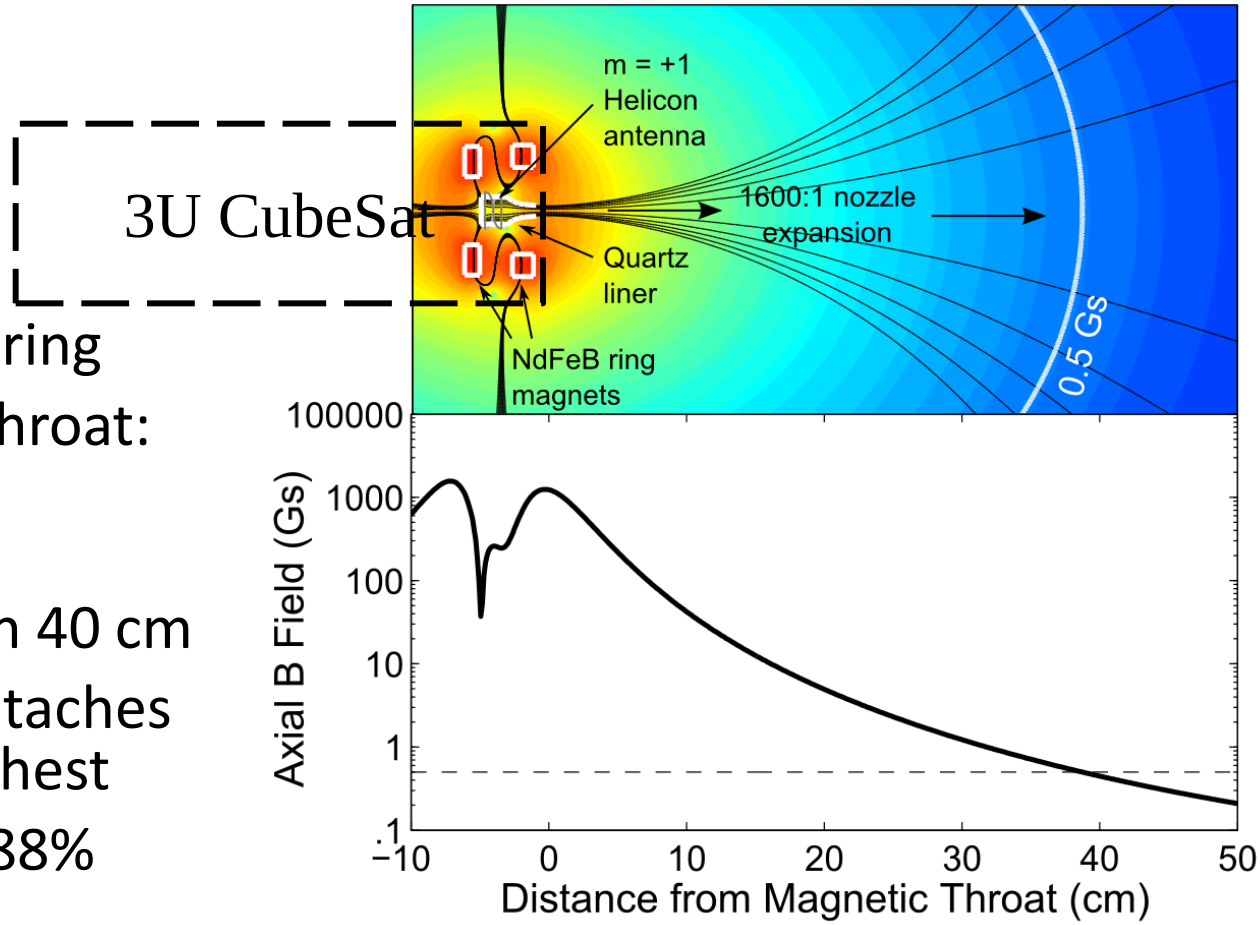
- Quartz plasma liner with showerhead
- Converging nozzle matches magnetic field
- C/D magnetic nozzle using permanent magnets
- Silver RF antenna for plasma generation/heating
- Faraday cage to isolate RF
- Fits in 1U CubeSat form factor



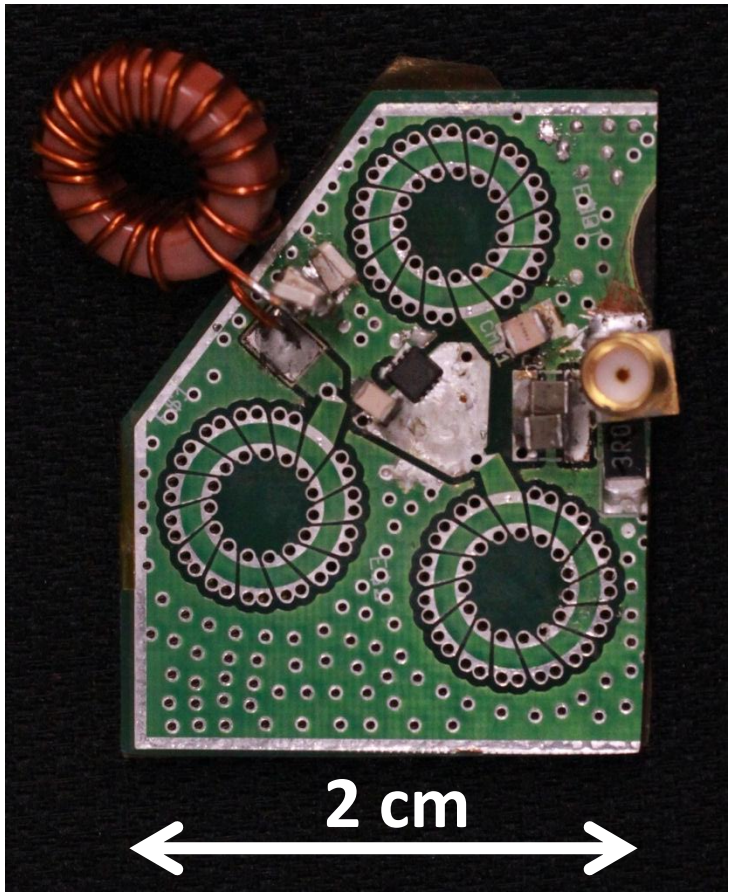


Permanent magnet nozzle

- NdFeB permanent magnets, N52
- Rings divided into radially oriented arc magnets for ease of manufacturing
- Magnetic field at throat: 1100 G
- Decays to Earth's magnetic field w/in 40 cm
- Assume plasma detaches at 0.5 G at the furthest
- Nozzle efficiency: 88%



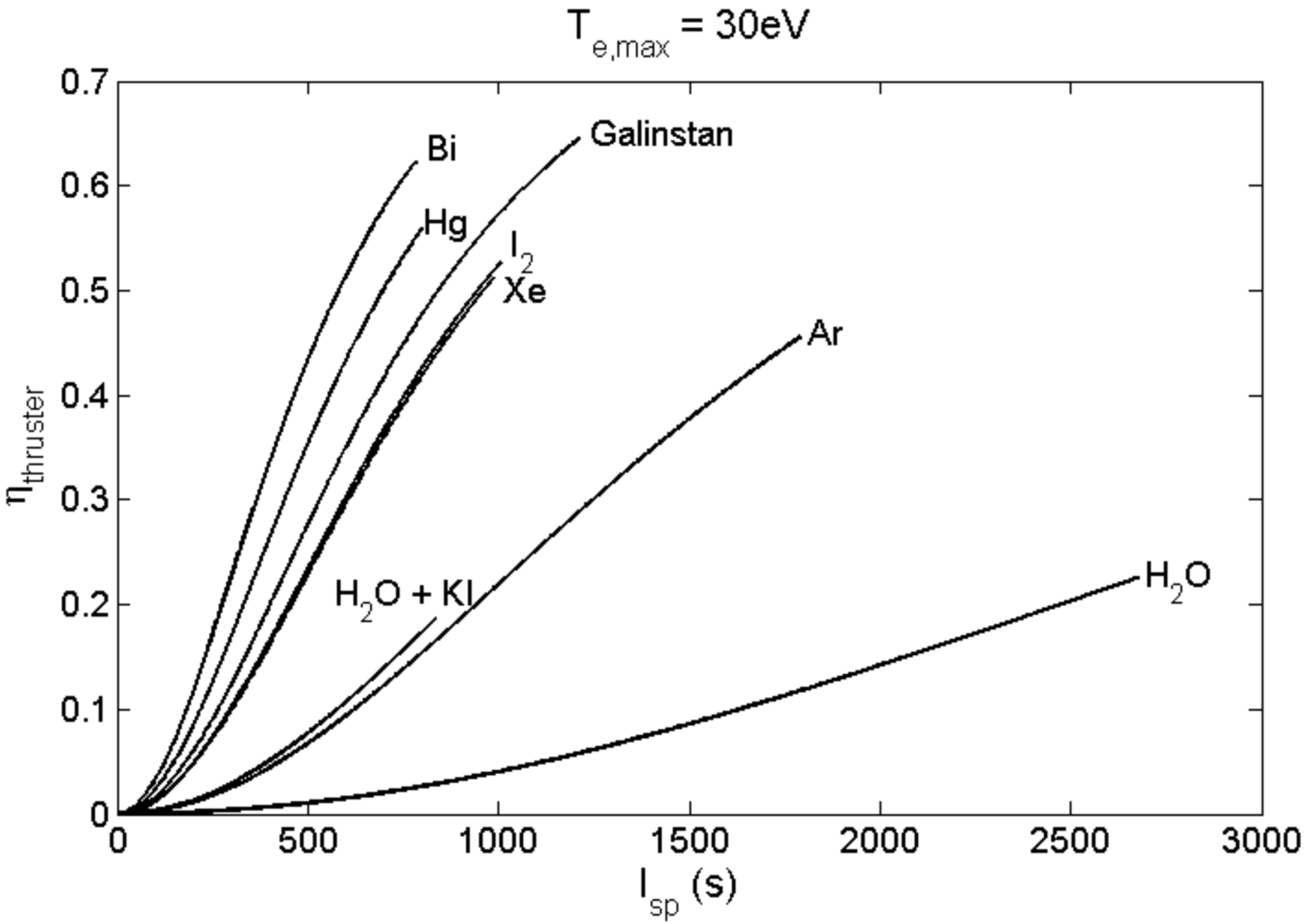
CAT: Prototype parameters



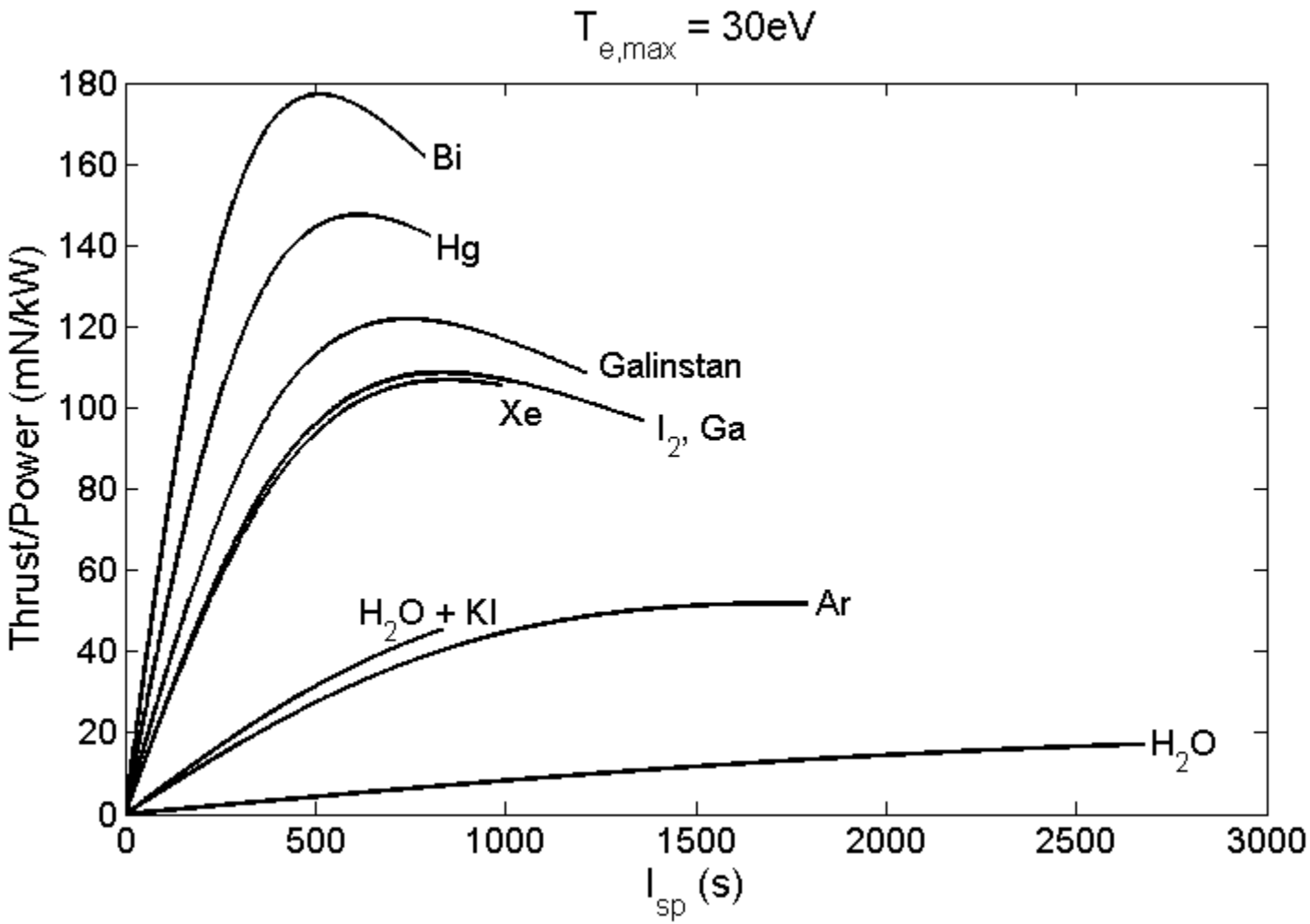
The operational prototype RF power processing unit

- 10 Watts (100 W pulsed)
- 27.13 MHz
- Propellant
  - Argon, Xenon: initial testing
  - Liquid metals (Iodine, Gallium, Galinstan)
    300C, 30C, -20 C
  - Water
- 0.1 – 1 sccm (0.1 to 4 mg/s)
- ~1100 G maximum B-field
- Magnetic nozzle field will decrease to strength of Earth's before reaching vacuum chamber wall, ~40 cm

Thruster performance from semi-empirical model

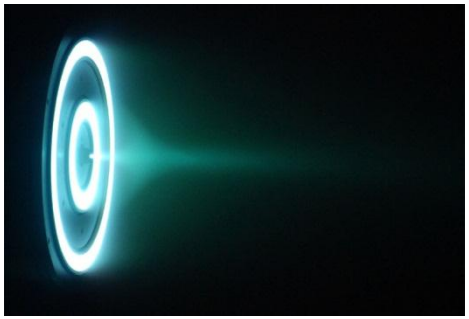


CAT design could be scaled up for moderately sized satellites



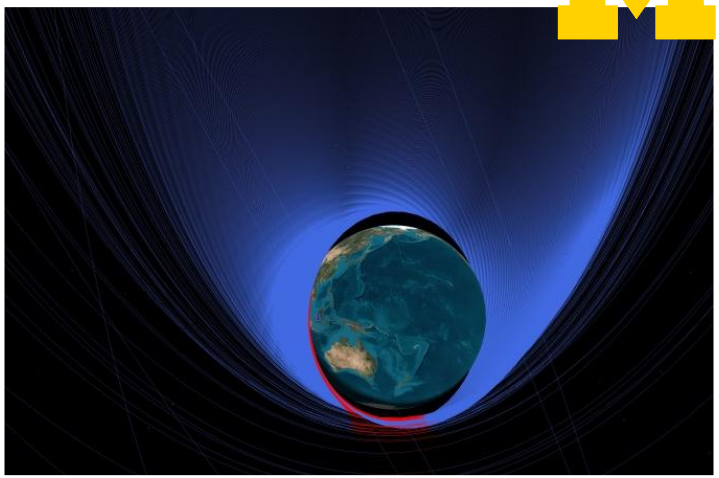
U. Michigan PEPL facilities afford us the most realistic testing before spaceflight

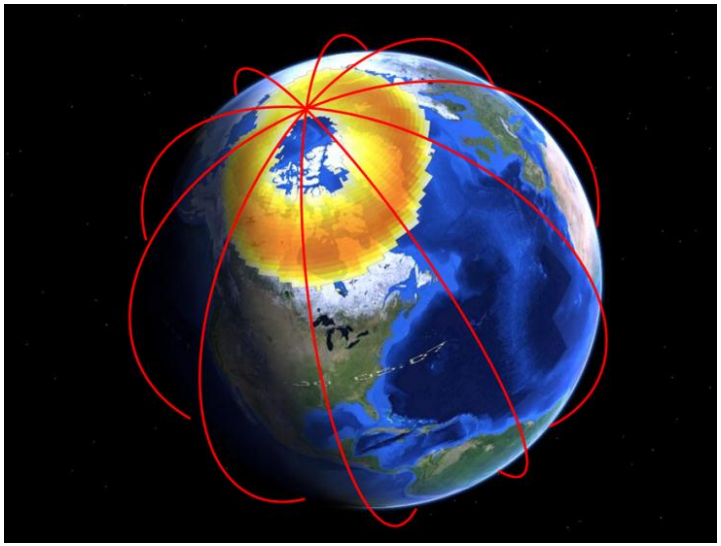
- 250 cubic meters
- 10^{-7} Torr
- 500,000 L/s pumping



New Missions Enabled

- Previously inaccessible orbits
  - Highly elliptical orbits
  - Geostationary orbits
  - Polar orbits
  - Earth-Moon, Earth-Sun Lagrange points
  - Earth escape from LEO, +3 km/s
- Cluster formation flying
- Long-lived low altitude orbits
- Cadre deployed from single LV

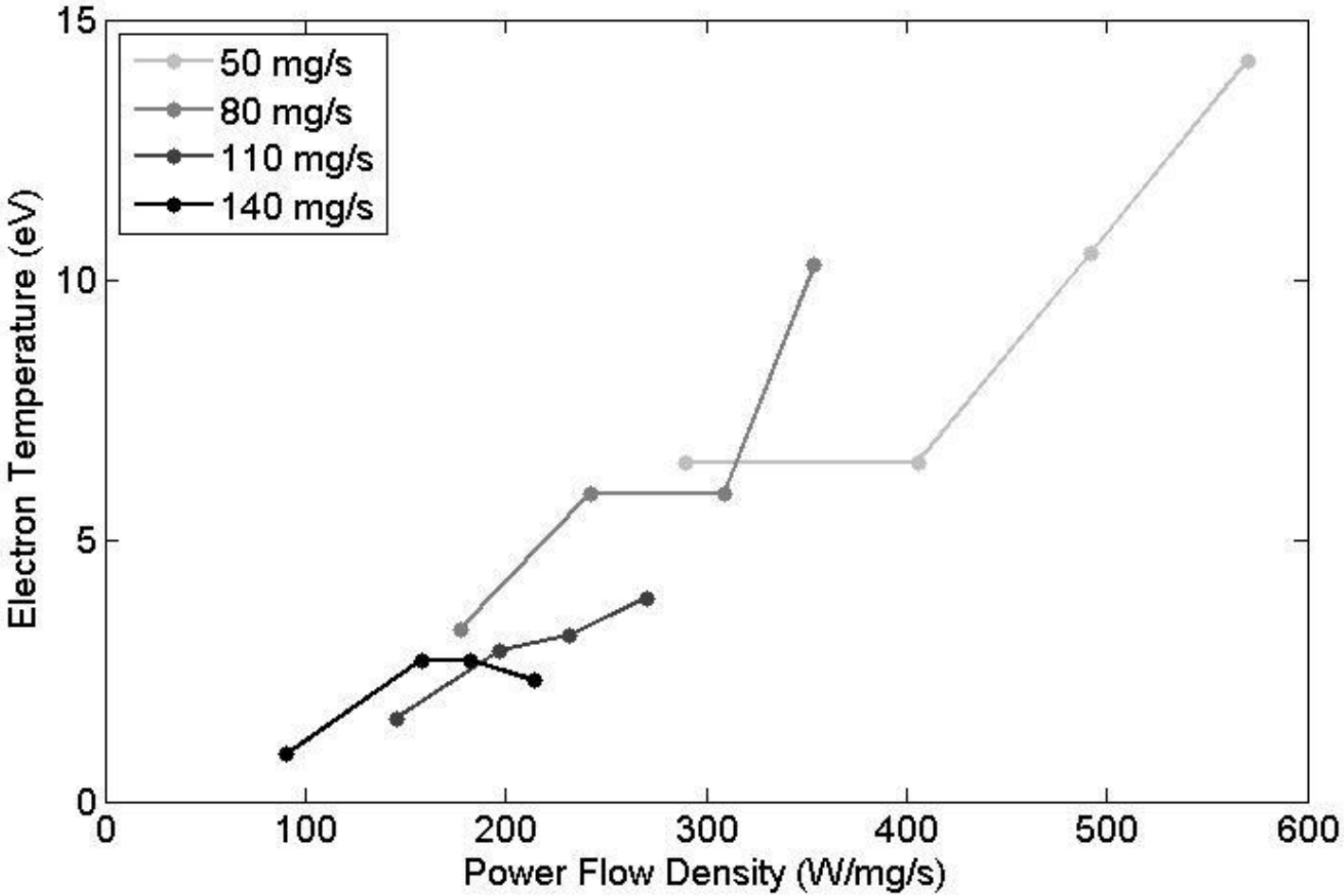




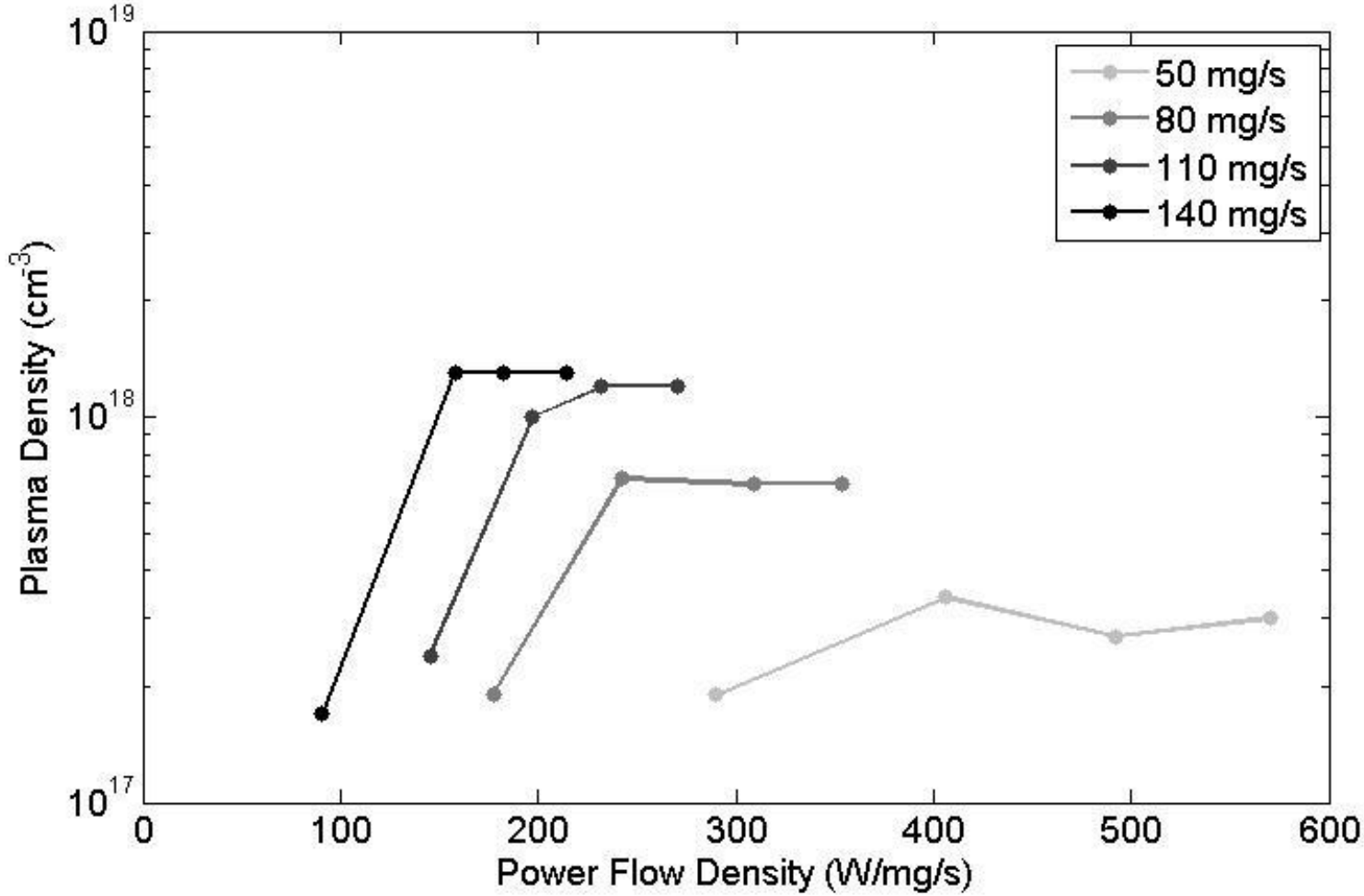
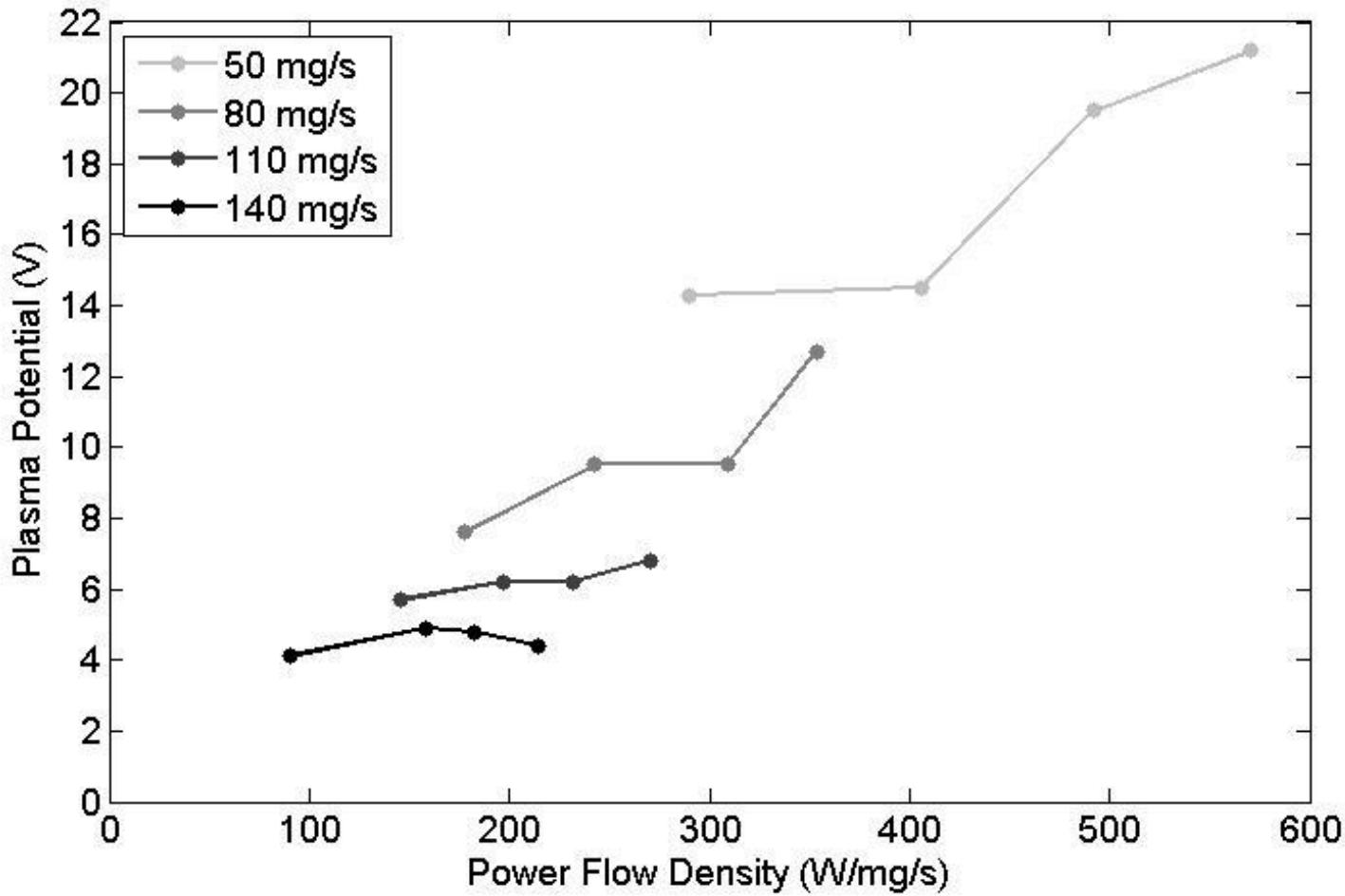
Conclusions

- (we hope that) CubeSats can soon expand our extraterrestrial mission capabilities
- The CubeSat Ambipolar Thruster (CAT) is a novel helicon thruster designed specifically for small satellites
  - Takes advantage of new insights on expanding plasmas in free space
  - Liquid storable propellants vastly reduce tank volume and mass
  - High thrust-to-power, up to 150 mN/kW
  - High delta-V, up to 10 km/s in 1U
  - Novel physics experiments needed in orbit
    - Science pull and technology push

Electron density increases with power flow density and mass flow



Density increases with power flow density, decreases with high mass flow



Plasma potential increases with power flow density